mergencystretch

Chapter 2. Identification and Adjustment

Interventions and the truncated factorization, confounding, the back-door and adjustment criteria

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2.1 Identification of causal effects

Motivation. The central question is how an SCM defines the distribution of a population under an intervention, and when that interventional distribution can be computed from the causal diagram and the observational distribution alone.

2.1.1 Structural causal models, revisited

Motivation. Recall the framework and introduce the do(·) operator.

2.1.1.1 Observation versus intervention

Reality is an unknown black-box system. We learn about it in two ways: *observation* (observational studies, passive records) and *intervention* (clinical trials, experiments). Questions such as drug efficacy or the effect of a hypothetical public policy concern the system *under intervention*, even when only observational data are available.

An SCM $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$ (unobserved variables, observed variables, mechanisms for $\mathbf{V}$, distribution of $\mathbf{U}$) is a mathematical understanding of the unknown reality. $\mathbf{V}$ are measured, $\mathbf{U}$ are unknown, and each measured variable is determined by a function of some measured and some unmeasured variables, $v \leftarrow f(\cdot)$. $\mathbf{U}$ can be viewed as what each unit actually is; $\mathbf{U}$ determines $\mathbf{V}$.

The SCM induces a causal diagram $\mathcal{G}$ and a whole family of distributions:

$$\begin{aligned} \mathcal{M} &\longmapsto \mathcal{G}, \\ \mathcal{M} &\longmapsto \{P,\ P_{x_1},\ P_{x_2},\ P_{x_2,x_4}, \dots\} \end{aligned}$$

corresponding to $\text{do}(\emptyset), \text{do}(x_1), \text{do}(x_2), \text{do}(x_2, x_4), \dots$

Layer 1 (observational). As derived in Chapter 1,

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{v}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \dots, v_{i-1}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).$$

The model also induces Layer 2 (interventional) and Layer 3 (counterfactual) distributions.

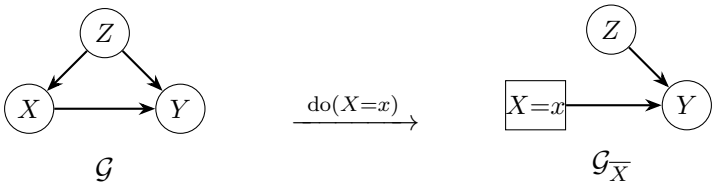
Causal diagram. Nodes for $\mathbf{V}$; a directed edge $B \to V_i$ when B is an argument of f_i (causal relationship); a bidirected edge $V_i \leftrightarrow V_j$ when $\mathbf{U}_i$ and $\mathbf{U}_j$ correlate (unobserved confounding).

2.1.1.2 The do-operator

Definition (Intervention)

Given $\mathcal{M}$, fixing $X \in \mathbf{V}$ to a constant x is written $\text{do}(X = x)$. It transforms $\mathcal{M}$ into the *submodel* $\mathcal{M}_x$, i.e. $\mathcal{M}$ with $f_X \in \mathcal{F}$ replaced by the constant x.

Graphically, the diagram of $\mathcal{M}_x$ is $\mathcal{G}_{\overline{X}}$: $\mathcal{G}$ with all edges *into* X removed.



$\mathcal{G}$ $\qquad$ $\mathcal{G}_{\overline{X}}$

2.1.2 The second layer: causal effects

Motivation. Define causal effects formally and compute them in Markovian models.

2.1.2.1 Connecting different worlds

Data come from P; the query concerns P', the distribution after a change. For instance, would fewer people get cancer if smoking were banned? The query $Q = P(\text{Cancer} = \text{true} \mid \text{do}(\text{Smoking} = \text{no}))$ is *not* an aspect of P.

Observation 1. The distribution alone tells us nothing about change; it describes static conditions of a population under a specific regime.
Observation 2. We must represent change: how the population reacts when it undergoes a change in regime.

The challenge. Let Z be age/sex, X an action, W a mediator, Y an outcome, with $Z \to X \to W \to Y$ and $Z \to Y$. In the hypothetical world X is set to x, cutting $Z \to X$. The real world generates $P(z, x, w, y)$; the hypothetical world generates $P(z, w, y \mid \text{do}(x))$. The goal is to know how much Y changes with X if we vary X between two different constants, free of the influence of Z; such variations are called *causal effects*. What is the relationship between $P(z, x, w, y)$ and $P(y \mid \text{do}(x))$? Is $P(y \mid \text{do}(x)) = P(y \mid x)$?

Definition (Causal effect; Pearl 2000, Def. 3.2.1)

Given disjoint sets $\mathbf{X}, \mathbf{Y}$, the causal effect of $\mathbf{X}$ on $\mathbf{Y}$, $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$, is a function from $\mathfrak{X}_{\mathbf{X}}$ to the space of distributions over $\mathbf{Y}$. For each $\mathbf{x}$, $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ is the probability of $\mathbf{Y} = \mathbf{y}$ induced by deleting from the model all equations of the variables in $\mathbf{X}$ and substituting $\mathbf{X} = \mathbf{x}$ in the remaining equations:

$$P(\mathbf{y} \mid \text{do}(\mathbf{x})) = P_{\mathbf{x}}(\mathbf{y}), \quad \text{the probability of } \mathbf{Y} = \mathbf{y} \text{ in } \mathcal{M}_{\mathbf{x}}.$$

2.1.2.2 Computing causal effects from observational data (Markovian case)

In the example, the observational and post-interventional distributions factorize as

$$P(\mathbf{v}) = P(z)\,P(x \mid z)\,P(w \mid x)\,P(y \mid w, z),$$
$$P_x(\mathbf{v} \setminus \{X\}) = P(z) \cdot \underbrace{1}_{X=x \text{ in } \mathcal{M}_x} \cdot P(w \mid x)\,P(y \mid w, z).$$

All other mechanisms are untouched by the intervention (*modularity*), so their conditional distributions are the same in both worlds. Hence

$$P(y \mid \text{do}(x)) = \sum_{z,w} P(z)P(w \mid x)P(y \mid w, z) \;\neq\; \sum_{z,w} P(z \mid x)P(w \mid x)P(y \mid w, z) = P(y \mid x)$$

in general: conditioning weights z by $P(z \mid x)$, intervening by $P(z)$.

Example 2.1 (Sprinkler: seeing versus doing)

Season $\to$ Sprinkler, Season $\to$ Rain, Sprinkler $\to$ Wet $\leftarrow$ Rain, Wet $\to$ Slippery.

$$Q_1 = P(\text{wt} \mid \text{Sp} = \text{on}) = \frac{\sum_{\text{sn,rn}} P(\text{sn})P(\text{Sp} = \text{on} \mid \text{sn})P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})}{\sum_{\text{sn}} P(\text{Sp} = \text{on} \mid \text{sn})P(\text{sn})},$$

$$Q_2 = P(\text{wt} \mid \text{do}(\text{Sp} = \text{on})) = \frac{\sum_{\text{sn,rn}} P(\text{sn})\, P'(\text{Sp} = \text{on})\, P(\text{rn} \mid \text{sn}) P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})}{\sum_{\text{sn}} P'(\text{Sp} = \text{on}) P(\text{sn})}$$
$$= \sum_{\text{sn,rn}} P(\text{sn}) P(\text{rn} \mid \text{sn}) P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn}),$$

where $P'(\text{Sp} = \text{on}) = 1$ is the interventional probability. Seeing the sprinkler on is evidence about the season and therefore about rain; turning it on is not.

2.1.3 Truncated factorization

Motivation. State the general formula for interventional distributions in Markovian models and interpret it as reweighting.

Theorem 2.1 (Truncated factorization; manipulation theorem, g-computation)

In a Markovian model, the distribution generated by $\text{do}(\mathbf{X} = \mathbf{x})$ is

$$P(\mathbf{v} \mid \text{do}(\mathbf{x})) = \begin{cases} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i) & \text{if } \mathbf{v} \text{ is consistent with } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases}$$

Derivation [Supplement]. $\mathcal{M}_{\mathbf{x}}$ is again Markovian, with the same mechanisms f_i and $P(u_i)$ for $V_i \notin \mathbf{X}$ and constant mechanisms for $\mathbf{X}$. Its Bayesian factorization (Chapter 1) is

$$\prod_{V_i \notin \mathbf{X}} P_{\mathbf{x}}(v_i \mid \mathbf{pa}_i) \prod_{X \in \mathbf{X}} \mathbb{1}[x \text{ agrees with } \mathbf{x}].$$

Since f_i and U_i are unchanged and $U_i \perp\!\!\!\perp Pa_i$ in both models,

$$P_{\mathbf{x}}(v_i \mid \mathbf{pa}_i) = \sum_{u_i} \mathbb{1}[f_i(\mathbf{pa}_i, u_i) = v_i] P(u_i) = P(v_i \mid \mathbf{pa}_i)$$

whenever the latter is defined (positivity). ■

Reweighting form. Since $P(\mathbf{v}) = \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i) \cdot \prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X)$,

$$P(\mathbf{v}' \mid \text{do}(\mathbf{x})) = \frac{P(\mathbf{v})}{\prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X)}, \qquad \mathbf{v}' = \mathbf{v} \setminus \mathbf{x}.$$

For a singleton X, with $\mathbf{v}'' = \mathbf{v} \setminus (\mathbf{pa}_X \cup \{x\})$,

$$P(\mathbf{v}' \mid \text{do}(x)) = \frac{P(\mathbf{v})}{P(x \mid \mathbf{pa}_X)} = \frac{P(\mathbf{v}) P(\mathbf{pa}_X)}{P(x, \mathbf{pa}_X)} = P(\mathbf{v}'' \mid x, \mathbf{pa}_X)\, P(\mathbf{pa}_X).$$

The passage from observational to interventional distribution is a *reweighting*: units with $X = x$ are up-weighted by $1/P(x \mid \mathbf{pa}_X)$ (the origin of inverse probability weighting, Chapter 6).

2.1.4 The identification problem

Motivation. Make precise when a causal query is determined by what we can observe and assume.

Definition (Identifiability; Pearl 2000, Def. 3.2.2)

The causal effect of $\mathbf{X}$ on $\mathbf{Y}$ is *identifiable* from $\mathcal{G}$ if $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ can be computed uniquely from any positive distribution over the observed variables; that is, for every pair of models $\mathcal{M}_1, \mathcal{M}_2$ inducing $\mathcal{G}$,

$$P^{\mathcal{M}_1}(\mathbf{v}) = P^{\mathcal{M}_2}(\mathbf{v}) > 0 \implies P^{\mathcal{M}_1}(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = P^{\mathcal{M}_2}(\mathbf{y} \mid \mathrm{do}(\mathbf{x})).$$

Picture. The true SCM $\mathcal{M}$ is unobserved; it induces the causal graph $\mathcal{G}$ (assumed known), the observational distribution $P(\mathbf{v})$ (observed) and the experimental distribution $P(\mathbf{v} \mid \mathrm{do}(\mathbf{x}))$ (unobserved). Among all models, consider those inducing $P(\mathbf{V})$ and those inducing $\mathcal{G}$. The query is identifiable iff it is constant over the intersection of these two sets: no matter the form of $\mathcal{M}_1, \mathcal{M}_2$, all models agreeing on $\mathcal{G}$ and $P(\mathbf{v})$ agree on $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$. If two models in the intersection disagree, the query is not identifiable.

Example 2.2 (A non-identifiable effect [Supplement])

Let $\mathcal{G}$ be $X \to Y$ with $X \leftrightarrow Y$, binary variables, $U \sim \mathrm{Bern}(1/2)$ shared. $\mathcal{M}_1$: $X \leftarrow U$, $Y \leftarrow U$ (no effect). $\mathcal{M}_2$: $X \leftarrow U$, $Y \leftarrow X$. Both give $P(X = Y = 0) = P(X = Y = 1) = 1/2$, yet $P^{\mathcal{M}_1}(Y = 1 \mid \mathrm{do}(X = 1)) = 1/2$ and $P^{\mathcal{M}_2}(Y = 1 \mid \mathrm{do}(X = 1)) = 1$. (Both are compatible with $\mathcal{G}$, since a graph only permits, never requires, dependence.)

2.1.4.1 Identification in Markovian models

Theorem 2.2

Given the causal diagram $\mathcal{G}$ of any Markovian model (all relevant variables measured), $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable for all disjoint $\mathbf{X}, \mathbf{Y}$, and

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i),$$

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{v}' \setminus \mathbf{y}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i).$$

Proof [Supplement]. By the truncated factorization theorem above, $P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i)$ evaluated at $\mathbf{X} = \mathbf{x}$. Each factor is a conditional distribution of the observational $P(\mathbf{v})$; under positivity ($P(\mathbf{pa}_i) > 0$ wherever needed) it is uniquely determined by $P(\mathbf{v})$. Marginalizing over $\mathbf{V} \setminus (\mathbf{X} \cup \mathbf{Y})$ gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ as a function of $P(\mathbf{v})$ alone. Hence any two Markovian models with diagram $\mathcal{G}$ that agree on $P(\mathbf{v})$ agree on $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$, which is the definition of identifiability. ■

The right-hand side involves only conditionals of $P(\mathbf{v})$, which are fixed by the observational distribution under positivity.

2.1.4.2 Adjustment by direct parents (singleton)

Theorem 2.3

In a Markovian model where only a subset of variables is measured, $Q = P(\mathbf{y} \mid \mathrm{do}(x))$ is identifiable whenever X, $\mathbf{Y}$ and Pa_X are measured, and

$$P(\mathbf{y} \mid \mathrm{do}(x)) = \sum_{\mathbf{pa}_X} P(\mathbf{y} \mid x, \mathbf{pa}_X) P(\mathbf{pa}_X).$$

Proof [Supplement]. From the singleton reweighting identity, $P(\mathbf{v}' \mid \mathrm{do}(x)) = P(\mathbf{v}'' \mid x, \mathbf{pa}_X)P(\mathbf{pa}_X)$; summing over $\mathbf{v}''\setminus\mathbf{y}$ gives the claim. Every unmeasured variable is summed out inside $P(\cdot \mid x, \mathbf{pa}_X)$, so only measured variables remain. ■

2.1.4.3 Adjustment by direct parents for a set of treatments (optional)

For $\mathbf{X} = \{X_1, \ldots, X_k\}$ adjustment by parents is not automatic: a parent of X_j may be a descendant of an earlier X_i. Notation:

$Pa_{\mathbf{X}} = \bigcup_{X\in\mathbf{X}} Pa_X$	parents of $\mathbf{X}$ (may intersect $\mathbf{X}$),
$Pa^{-}_{\mathbf{W}} = Pa_{\mathbf{W}} \setminus \mathbf{W}$	parents of $\mathbf{W}$ excluding $\mathbf{W}$,
$\mathbf{X}_{<i} = \{X_1, \ldots, X_{i-1}\}$, $\mathbf{X}_{\le i}$	predecessors in a topological order,
$Pa'_{X_i} = Pa_{X_i} \setminus (\mathbf{X}_{<i} \cup Pa^{-}_{\mathbf{X}_{<i}})$	"new" parents of X_i.

Theorem 2.4

Let $\mathbf{X}$ be topologically ordered. If no element of Pa'_{X_j} is a descendant of X_i for every $i < j$, then

$$\prod_{X\in\mathbf{X}} P(x \mid \mathbf{pa}_X) = P(\mathbf{x} \mid \mathbf{pa}^{-}_{\mathbf{X}}),$$

hence

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{pa}^{-}_{\mathbf{X}}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}^{-}_{\mathbf{X}})P(\mathbf{pa}^{-}_{\mathbf{X}}).$$

Proof [Supplement]. By induction we show $\prod_{X\in\mathbf{X}_{\le i}} P(x \mid \mathbf{pa}_X) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}})$.

Base case. $P(x_1 \mid \mathbf{pa}_{X_1}) = P(x_1 \mid \mathbf{pa}_{X_1} \setminus \{X_1\}) = P(x_1 \mid \mathbf{pa}^{-}_{X_1}) = P(\mathbf{x}_{\le 1} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le 1}})$.

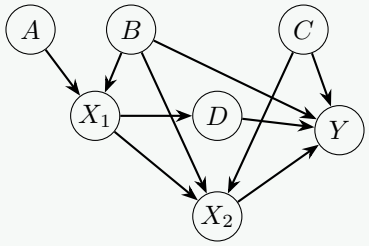
Step. Assume the claim for $i-1$. It suffices to show $P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}_{X_i}) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}})$. Note $Pa^{-}_{\mathbf{X}_{\le i}} = Pa^{-}_{\mathbf{X}_{<i}} \cup Pa'_{X_i}$. Then

$$\begin{aligned}
&P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}_{X_i}) \\
&= P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}'_{X_i}, \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) && \text{Markov condition} \\
&= P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}'_{X_i}, \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})\frac{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \text{multiply by 1} \\
&= \frac{P(\mathbf{pa}'_{X_i}, \mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \text{chain rule, } \{X_i\}\cup\mathbf{X}_{<i} = \mathbf{X}_{\le i} \\
&= \frac{P(\mathbf{pa}'_{X_i}, \mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \mathbf{X}_{<i} \perp\!\!\!\perp Pa'_{X_i} \mid Pa^{-}_{\mathbf{X}_{<i}} \\
&= P(\mathbf{x}_{\le i} \mid \mathbf{pa}'_{X_i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}}).
\end{aligned}$$

The first line replaces the parents of X_i by a superset of non-descendants (local Markov: $\mathbf{X}_{<i}$ and $Pa^{-}_{\mathbf{X}_{<i}}$ are non-descendants of X_i since they precede it or are parents of predecessors). The key independence holds by d-separation: Pa'_{X_i} are non-descendants of every X_j, $j < i$, so every path from $\mathbf{X}_{<i}$ to Pa'_{X_i} either enters X_j through a parent (blocked by $Pa^{-}_{\mathbf{X}_{<i}}$) or leaves X_j along a directed edge, in which case it must meet a collider that is not an ancestor of the conditioning set. ■

Example 2.3

Graph: $A \to X_1$, $B \to X_1$, $B \to X_2$, $B \to Y$, $C \to X_2$, $C \to Y$, $X_1 \to D$, $X_1 \to X_2$, $D \to Y$, $X_2 \to Y$.



Here $Pa_{X_1} = \{A, B\}$, $Pa_{X_2} = \{X_1, B, C\}$, $Pa^-_{\mathbf{X}} = \{A, B, C\}$ and $Pa'_{X_2} = \{C\}$, which is not a descendant of X_1. First, $P(y \mid \mathrm{do}(\mathbf{x})) = \sum P(\mathbf{v})/[P(x_1 \mid a, b)P(x_2 \mid x_1, b, c)]$. Focusing on the denominator,

$$\begin{aligned} P(x_1 \mid a,b)P(x_2 \mid x_1,b,c) &= P(x_1 \mid a,b)P(x_2 \mid \underbrace{x_1}_{\mathbf{X}_{<2}}, \underbrace{a,b}_{Pa^-_{\mathbf{X}_{<2}}}, \underbrace{c}_{Pa'_{X_2}}) \\ &= P(x_1 \mid a,b)P(x_2 \mid x_1,a,b,c)\frac{P(c \mid x_1,a,b)}{P(c \mid x_1,a,b)} \\ &= \frac{P(x_1,x_2,c \mid a,b)}{P(c \mid x_1,a,b)} = \frac{P(x_1,x_2,c \mid a,b)}{P(c \mid a,b)} \qquad = P(x_1,x_2 \mid a,b,c) = P(\mathbf{x} \mid \mathbf{pa}^-_{\mathbf{X}}), \end{aligned}$$

so $P(y \mid \mathrm{do}(x_1, x_2)) = \sum_{a,b,c} P(y \mid x_1, x_2, a, b, c)P(a, b, c)$. D (a descendant of X_1) is correctly *not* adjusted for.

2.1.4.4 When a parent is latent

In the sprinkler model, suppose Season is unobserved. Is Q_2 still computable?

$$\begin{aligned} Q_2 &= \sum_{\mathrm{sn,rn}} P(\mathrm{sn})P(\mathrm{rn} \mid \mathrm{sn})P(\mathrm{wt} \mid \mathrm{Sp} = \mathrm{on}, \mathrm{rn}) \\ &= \sum_{\mathrm{sn,rn}} P(\mathrm{rn}, \mathrm{sn})P(\mathrm{wt} \mid \mathrm{Sp} = \mathrm{on}, \mathrm{rn}) \\ &= \sum_{\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{Sp} = \mathrm{on}, \mathrm{rn})P(\mathrm{rn}). \end{aligned}$$

Yes: adjusting for Rain instead of the latent parent Season suffices. Adjustment sets other than the parents can work; characterizing them is the subject of the back-door criterion (Section 2.2).

2.1.5 Summary

- Causal effects $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ are defined through submodels (deleting and substituting equations).
- In Markovian models every causal effect is identifiable via the truncated factorization.
- Adjustment by direct parents gives a practical formula: condition on the parents of the treatment and average over their marginal distribution.
- The identification problem asks whether $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is uniquely determined by $\mathcal{G}$ and $P(\mathbf{v})$.
- With latent variables identification becomes non-trivial; this continues with the back-door criterion and beyond.

2.2 Confounding and the back-door criterion

Motivation. The central question is how a causal effect can be identified when some parents of the treatment are unobserved, and how the resulting adjustment formula can be evaluated from finite samples.

2.2.1 Recap: identifiable and non-identifiable effects

Motivation. Illustrate both sides of the identifiability definition with concrete pairs of models.

2.2.1.1 An identifiable effect

Take $\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to Y$ (no latent confounding). Any compatible model factorizes as

$$\begin{aligned} P(\mathbf{v}) &= P(z)P(x \mid z)P(y \mid x, z), \\ P(\mathbf{v}' \mid \mathrm{do}(x)) &= P(z)P(y \mid x, z), \\ P(y \mid \mathrm{do}(x)) &= \sum_z P(z)P(y \mid x, z). \end{aligned}$$

Consider two models

$$\mathcal{M}^{(j)} = \begin{cases} Z \leftarrow f_z^{(j)}(u_z) \\ X \leftarrow f_x^{(j)}(z, u_x) \\ Y \leftarrow f_y^{(j)}(x, z, u_y) \end{cases}$$

$$\mathcal{M}_x^{(j)} = \begin{cases} Z \leftarrow f_z^{(j)}(u_z) \\ X \leftarrow x \\ Y \leftarrow f_y^{(j)}(x, z, u_y) \end{cases}$$

$$j = 1, 2.$$

Whatever the functions and $P(\mathbf{u})$, if $\mathcal{M}^{(1)}, \mathcal{M}^{(2)}$ agree on $\langle \mathcal{G}, P(\mathbf{v}) \rangle$ they agree on $P(z)$ and $P(y \mid x, z)$, hence on $P(\mathbf{v}' \mid \mathrm{do}(x))$.

2.2.1.2 A non-identifiable effect

Take $\mathcal{G}$: $X \to Y$ and $X \leftrightarrow Y$ (through a latent U_{xy}). Let U_y, U_{xy} be independent fair coins in both models and $X \leftarrow U_{xy}$ in the observational world. Define

$$\mathcal{M}^{(1)} : \; Y \leftarrow (X \oplus U_{xy}) \vee U_y, \qquad \mathcal{M}^{(2)} : \; Y \leftarrow U_y.$$

Observationally $X = U_{xy}$, so $X \oplus U_{xy} = 0$ and $Y = U_y$ in both models: the observational distributions coincide. Under $\mathrm{do}(x)$:

U_y	U_{xy}	X	$Y^{(1)}$	$Y^{(2)}$	$P(\mathbf{u})$
0	0	x	x	0	1/4
0	1	x	$\neg x$	0	1/4
1	0	x	1	1	1/4
1	1	x	1	1	1/4

Y	$P^{(1)}(Y \mid \mathrm{do}(x))$	$P^{(2)}(Y \mid \mathrm{do}(x))$
0	1/4	1/2
1	3/4	1/2

In $\mathcal{M}^{(1)}$, $Y = 0$ requires $U_y = 0$ and $x = U_{xy}$, probability 1/4. Both models induce $\mathcal{G}$ and the same $P(\mathbf{v})$, yet $P^{(1)}(y \mid \mathrm{do}(x)) \neq P^{(2)}(y \mid \mathrm{do}(x))$. In general $P(\mathbf{v} \mid \mathrm{do}(x)) = \sum_{u_{xy}} P(y \mid x, u_{xy})P(u_{xy})$ depends on the latent U_{xy}, which the data do not reveal.

2.2.2 Confounding bias

Motivation. Show how a common cause makes conditioning differ from intervening.

What is the causal effect of exercise on cholesterol? Age affects both (age → exercise, age → cholesterol, exercise → cholesterol). In the scatter plot, older people exercise more and also have higher cholesterol, so pooled data show cholesterol *increasing* with exercise, while within every age group more exercise means lower cholesterol. Hence

$$P(\text{cholesterol} \mid \text{exercise}) \neq P(\text{cholesterol} \mid \text{do(exercise)}).$$

This difference is **confounding bias**, one of the major obstacles to causal inference and interpretability.

Is confounding bias removable? Goal: find $Q = P(y \mid \text{do}(x))$ from measurements on $Z_1, \ldots, Z_k$ when some parents of X are unobserved. Adjustment by direct parents (Section 2.1) is then unavailable. Which subsets of measured variables can replace the parents?

2.2.3 The back-door criterion

Motivation. Give a graphical condition under which a covariate set can substitute for the parents of the treatment.

Definition (Back-door criterion; Pearl 2000, Def. 3.3.1)

A set $\mathbf{Z}$ satisfies the back-door criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ if

(i) no node in $\mathbf{Z}$ is a descendant of $\mathbf{X}$; and

(ii) $\mathbf{Z}$ blocks every path between $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ that contains an arrow into X (a *back-door path*).

Theorem 2.5 (Back-door adjustment)

If $\mathbf{Z}$ satisfies the back-door criterion relative to $(\mathbf{X}, \mathbf{Y})$, the effect is identifiable and

$$P(\mathbf{y} \mid \text{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

Proof [Supplement]. We use two rules of do-calculus (Chapter 3), which are proved there from the truncated factorization alone, so there is no circularity. By the probability axioms,

$$P(\mathbf{y} \mid \text{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \text{do}(\mathbf{x}), \mathbf{z})\, P(\mathbf{z} \mid \text{do}(\mathbf{x})).$$

Step 1. $P(\mathbf{z} \mid \text{do}(\mathbf{x})) = P(\mathbf{z})$. In $\mathcal{G}_{\overline{\mathbf{X}}}$ every edge at $\mathbf{X}$ points out of $\mathbf{X}$. A path from $\mathbf{X}$ to $\mathbf{Z}$ without colliders would then be directed from $\mathbf{X}$ to $\mathbf{Z}$, making an element of $\mathbf{Z}$ a descendant of $\mathbf{X}$, which condition (i) excludes. Every other path contains a collider; since we condition on nothing, the path is blocked. So $(\mathbf{Z} \perp_d \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}}$, and Rule 3 removes the action.

Step 2. $P(\mathbf{y} \mid \text{do}(\mathbf{x}), \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})$. In $\mathcal{G}_{\underline{\mathbf{X}}}$ the edges out of $\mathbf{X}$ are deleted, so every path between $\mathbf{X}$ and $\mathbf{Y}$ begins with an arrow into $\mathbf{X}$: it is a back-door path, and condition (ii) says $\mathbf{Z}$ blocks it. Thus $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z})_{\mathcal{G}_{\underline{\mathbf{X}}}}$ and Rule 2 exchanges the action for an observation.

Step 3. Substituting both identities gives $P(\mathbf{y} \mid \text{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$. ■

2.2.3.1 Back-door sets substitute for the parents: an example

In the sprinkler network, Rain satisfies the criterion relative to (Sprinkler, Wet):

(i) Rain is not a descendant of Sprinkler;

(ii) Rain blocks the only back-door path Sprinkler ← Season → Rain → Wet. Starting from adjustment for the direct parent Season,

$$\begin{aligned} P(\mathrm{wt} \mid \mathrm{do}(\mathrm{sp})) &= \sum_{\mathrm{sn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{sn}) P(\mathrm{sn}) \\ &= \sum_{\mathrm{sn},\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{sn}, \mathrm{rn}) P(\mathrm{rn} \mid \mathrm{sp}, \mathrm{sn}) P(\mathrm{sn}) \\ &= \sum_{\mathrm{sn},\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{rn}) P(\mathrm{rn} \mid \mathrm{sn}) P(\mathrm{sn}) \\ &= \sum_{\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{rn}) \sum_{\mathrm{sn}} P(\mathrm{rn}, \mathrm{sn}) \\ &= \sum_{\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{rn}) P(\mathrm{rn}). \end{aligned}$$

The third equality uses Wet ⫫ Season | Sprinkler, Rain and Rain ⫫ Sprinkler | Season, both read off by d-separation.

2.2.3.2 General proof: back-door adjustment = parent adjustment

Let $Pa_{\mathbf{X}}^{-}$ be the parents of $\mathbf{X}$ outside $\mathbf{X}$, and $\mathbf{Z}^{-} = \mathbf{Z} \setminus Pa_{\mathbf{X}}^{-}$. The two conditions imply two independences:

$$\begin{aligned} &\text{(i)} \implies \mathbf{X} \perp\!\!\!\perp (\mathbf{Z} \setminus Pa_{\mathbf{X}}^{-}) \mid Pa_{\mathbf{X}}^{-} && \text{(local Markov)}, \\ &\text{(ii)} \implies \mathbf{Y} \perp\!\!\!\perp (Pa_{\mathbf{X}}^{-} \setminus \mathbf{Z}) \mid \mathbf{Z}, \mathbf{X} && \text{(back-door paths blocked)}. \end{aligned}$$

The first holds because $\mathbf{Z}$ consists of non-descendants of $\mathbf{X}$; the second because every path from a parent of $\mathbf{X}$ to $\mathbf{Y}$ that avoids $\mathbf{X}$ extends to a back-door path. Then

$$\begin{aligned} P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) &= \sum_{\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}) P(\mathbf{pa}_{\mathbf{X}}^{-}) && \text{parent adjustment} \\ &= \sum_{\mathbf{z}^{-},\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}, \mathbf{z}^{-}) P(\mathbf{z}^{-} \mid \mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}) P(\mathbf{pa}_{\mathbf{X}}^{-}) && \text{def. of cond. probability} \\ &= \sum_{\mathbf{z}^{-},\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}^{-} \mid \mathbf{pa}_{\mathbf{X}}^{-}) P(\mathbf{pa}_{\mathbf{X}}^{-}) && \text{(ii) and (i)} \\ &= \sum_{\mathbf{z}^{-},\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}^{-}, \mathbf{pa}_{\mathbf{X}}^{-}) \\ &= \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) \sum_{\mathbf{pa}_{\mathbf{X}}^{-} \setminus \mathbf{z}} P(\mathbf{z}^{-}, \mathbf{pa}_{\mathbf{X}}^{-}) && \text{by algebra} \\ &= \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}). \end{aligned}$$

In the third line, $\{\mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}, \mathbf{z}^{-}\} = \{\mathbf{x}, \mathbf{z}\} \cup (\mathbf{pa}_{\mathbf{X}}^{-} \setminus \mathbf{z})$, so (ii) removes $\mathbf{pa}_{\mathbf{X}}^{-} \setminus \mathbf{z}$ from the conditioning set;

(i) removes $\mathbf{x}$ from $P(\mathbf{z}^{-} \mid \cdot)$. *Adjustment by* $\mathbf{Z}$ *is equivalent to adjustment by the direct parents whenever* $\mathbf{Z}$ *is back-door admissible.* (The parent formula holds in the Markovian model containing all variables, latent ones included; the final expression uses only measured variables.)

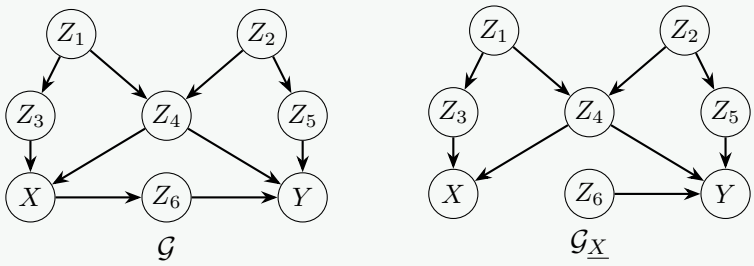
2.2.3.3 A graphical test

Condition (ii) says that $\mathbf{Z}$ d-separates $\mathbf{X}$ from $\mathbf{Y}$ once the causal paths are removed. Let $\mathcal{G}_{\underline{\mathbf{X}}}$ be $\mathcal{G}$ with all edges *out of* $\mathbf{X}$ deleted. Then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) \text{ is identified by } \mathbf{Z} \quad \text{if (i) and } (\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z})_{\mathcal{G}_{\underline{\mathbf{X}}}}.$$

Example 2.4 (Pearl's seven-variable model)

$Z_1 \to Z_3 \to X$, $Z_1 \to Z_4$, $Z_2 \to Z_4$, $Z_2 \to Z_5 \to Y$, $Z_4 \to X$, $Z_4 \to Y$, $X \to Z_6 \to Y$.



$\mathcal{G}$ $\mathcal{G}_{\underline{X}}$

The back-door paths are $X \leftarrow Z_4 \to Y$, $X \leftarrow Z_4 \leftarrow Z_2 \to Z_5 \to Y$, $X \leftarrow Z_3 \leftarrow Z_1 \to Z_4 \to Y$ and $X \leftarrow Z_3 \leftarrow Z_1 \to Z_4 \leftarrow Z_2 \to Z_5 \to Y$. Conditioning on Z_4 blocks the first three but opens the collider Z_4 on the last, which Z_1 blocks again. Hence $\{Z_1, Z_4\}$ is admissible:

$$P(y \mid \mathrm{do}(x)) = \sum_{z_1, z_4} P(y \mid x, z_1, z_4) P(z_1, z_4).$$

Other admissible sets include $\{Z_3, Z_4\}$, $\{Z_4, Z_2\}$, $\{Z_4, Z_5\}$; $\{Z_4\}$ alone is not admissible and Z_6 never is (descendant of X).

Example 2.5 (More back-door examples [Supplement])

(a) $Z_2 \to Z_4$, $Z_2 \to Z_5$, $Z_4 \to X$, $Z_4 \to Y$, $X \leftrightarrow Z_4$, $X \to Z_6 \to Y$, $Z_5 \to Y$. Z_4 must be included to block $X \leftarrow Z_4 \to Y$; but Z_4 is a collider on $X \leftrightarrow Z_4 \leftarrow Z_2 \to Z_5 \to Y$, so Z_2 or Z_5 must be added. Admissible: $\mathbf{Z} \in \{\{Z_4, Z_2\}, \{Z_4, Z_5\}, \{Z_4, Z_2, Z_5\}\}$.
(b) $X \to Z$, $X \to Y$. There is no back-door path and Z is a descendant of X, so $\mathbf{Z} = \emptyset$ and $P(y \mid \mathrm{do}(x)) = P(y \mid x)$.

2.2.4 Evaluating back-door adjustment: inverse probability weighting

Motivation. Turn the adjustment formula into an estimator whose cost does not explode with the dimension of $\mathbf{Z}$.

The criterion says when $\mathbf{Z}$ is admissible,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}),$$

$$\mathbb{E}[Y \mid \mathrm{do}(\mathbf{x})] = \mathbb{E}_{\mathbf{Z}}[\mathbb{E}[Y \mid \mathbf{x}, \mathbf{Z}]].$$

Evaluating it naively requires estimating many conditional distributions and summing over a possibly high-dimensional $\mathbf{Z}$, roughly $O(\exp(|\mathbf{Z}|))$ time, with sparse cells for each $\mathbf{z}$.

2.2.4.1 Rewriting with the propensity score

$$P(y \mid \mathrm{do}(x)) = \sum_{\mathbf{z}} P(y \mid x, \mathbf{z}) P(\mathbf{z}) = \sum_{\mathbf{z}} \frac{P(y, x, \mathbf{z})}{P(x, \mathbf{z})} P(\mathbf{z}) = \sum_{\mathbf{z}} \frac{P(y, x, \mathbf{z})}{P(x \mid \mathbf{z})}.$$

The numerator is an entry of the joint distribution; the denominator is the assignment probability (propensity), fitted by a model $g(\mathbf{z}) \approx P(x \mid \mathbf{z})$.

2.2.4.2 Sample version

With N samples, replace $P(y, x, \mathbf{z})$ by its empirical frequency:

$$\begin{aligned} P(y \mid \mathrm{do}(x)) &\approx \sum_{\mathbf{z}} \frac{\frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N}\sum_{i=1}^{N}\sum_{\mathbf{z}} \frac{\mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N}\sum_{i=1}^{N} \frac{\mathbb{1}(y_i = y, x_i = x)}{g(\mathbf{z}_i)}. \end{aligned}$$

The sum over $\mathbf{z}$ collapses because only $\mathbf{z} = \mathbf{z}_i$ contributes. The cost is proportional to N, not to $|\mathfrak{X}_{\mathbf{Z}}|$. ($\mathbb{1}$(predicate) is 1 if the predicate is true and 0 otherwise.) Similarly,

$$\begin{aligned} \mathbb{E}[Y \mid \mathrm{do}(x)] &= \sum_{\mathbf{z},y} y\, \frac{P(y, x, \mathbf{z})}{P(x \mid \mathbf{z})} \\ &\approx \frac{1}{N}\sum_{i=1}^{N}\sum_{\mathbf{z},y} \frac{y\,\mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N}\sum_{i=1}^{N} \frac{y_i\,\mathbb{1}(x_i = x)}{g(\mathbf{z}_i)}. \end{aligned}$$

2.2.4.3 Average treatment effect

For binary X with $g(\mathbf{z}) = P(X = 1 \mid \mathbf{z})$, when the back-door criterion holds,

$$\begin{aligned} \widehat{\mathrm{ATE}} &= \hat{\mathbb{E}}[Y \mid \mathrm{do}(X = 1)] - \hat{\mathbb{E}}[Y \mid \mathrm{do}(X = 0)] \\ &= \frac{1}{N}\sum_{i=1}^{N} y_i \left(\frac{\mathbb{1}(x_i = 1)}{g(\mathbf{z}_i)} - \frac{\mathbb{1}(x_i = 0)}{1 - g(\mathbf{z}_i)} \right) \\ &= \frac{1}{N}\sum_{i=1}^{N} \left(\frac{y_i x_i}{g(\mathbf{z}_i)} - \frac{y_i(1 - x_i)}{1 - g(\mathbf{z}_i)} \right). \end{aligned}$$

Notation varies across fields: covariates are often X or L, treatment T, W or A.

2.2.4.4 Interpretation: pseudo-causal samples

IPW turns observational samples into a "pseudo-population" that mimics a randomized experiment: samples with $x_i \neq x$ are discarded and each remaining sample is replicated $1/g(\mathbf{z}_i)$ times. A unit with $P(x_1 \mid \mathbf{z}_1) = 1/3$ counts three times, one with $P(x_3 \mid \mathbf{z}_3) = 1/2$ counts twice. Units that were unlikely to receive treatment given their covariates stand in for the many similar units that were not treated, removing the dependence between X and $\mathbf{Z}$.

2.2.5 Summary

- Confounding bias arises when treatment and outcome share a common cause, making $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$.

- The back-door criterion is a graphical test of whether $\mathbf{Z}$ is admissible for adjustment.
- Back-door adjustment: $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$.
- It generalizes adjustment by direct parents: any back-door admissible set can substitute for the parents.
- IPW evaluates the back-door expression from samples without summing over a high-dimensional $\mathbf{Z}$.

2.3 The adjustment criterion

Motivation. The central question is exactly which covariate sets $\mathbf{Z}$ make the adjustment formula $\sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$ equal to $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$, and how such sets can be tested, constructed and enumerated efficiently.

2.3.1 Back-door criterion: recap and two refinements

Motivation. Recall back-door adjustment and see how it resolves Simpson's paradox and conditional effects.

Theorem 2.6 (Back-door adjustment; Pearl 2000, Thm. 3.3.2)

If $\mathbf{Z}$ satisfies (i) no node in $\mathbf{Z}$ is a descendant of $\mathbf{X}$, and (ii) $\mathbf{Z}$ blocks every path between $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ containing an arrow into X, then $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$.

Proof [Supplement]. This is the back-door adjustment theorem of Section 2.2; the proof given there (Rule 3 for $P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}))$, Rule 2 for $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z})$) applies verbatim. ■

2.3.1.1 Back to Simpson's paradox

Recall the drug/sex/recovery tables of Chapter 1 and the three stories:

	$F \to X,\ F \to Y,\ X \to Y$	$X \to F \to Y,\ X \to Y$	$X \leftarrow U_1 \to F \leftarrow U_2 \to Y,\ X \to Y$
$\emptyset$	not admissible	admissible	admissible
$\{F\}$	admissible	not admissible ($F \in De(X)$)	not admissible (opens collider)
Use	F-specific tables	aggregated table	aggregated table

The back-door criterion turns the physician's dilemma into a mechanical check on the graph.

2.3.1.2 Conditional back-door

Theorem 2.7 (Conditional back-door criterion)

The $\mathbf{w}$-specific effect $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ is identified whenever we can measure a set $\mathbf{Z}$ such that $\mathbf{W} \cup \mathbf{Z}$ satisfies the back-door criterion. Moreover,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, \mathbf{w}) P(\mathbf{z} \mid \mathbf{w}).$$

Proof [Supplement]. By the probability axioms, $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$.

Step 1. Since $\mathbf{W} \cup \mathbf{Z}$ blocks every back-door path and all paths in $\mathcal{G}_{\underline{\mathbf{X}}}$ are back-door paths, $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z}, \mathbf{W})_{\mathcal{G}_{\underline{\mathbf{X}}}}$; Rule 2 gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, \mathbf{w})$.

Step 2. No element of $\mathbf{W}$ is a descendant of $\mathbf{X}$, so $\mathbf{X}(\mathbf{W}) = \mathbf{X}$ in Rule 3 and the relevant graph is $\mathcal{G}_{\overline{\mathbf{X}}}$. A path from $\mathbf{X}$ to $\mathbf{Z}$ there leaves $\mathbf{X}$ along an outgoing edge. If it has no collider, it is directed and $\mathbf{Z}$ would contain a descendant of $\mathbf{X}$. Otherwise its first collider C is reached by a directed path from $\mathbf{X}$, so C and all its descendants are descendants of $\mathbf{X}$, hence not in $\mathbf{W}$; the path is blocked given $\mathbf{W}$. Thus $(\mathbf{Z} \perp_d \mathbf{X} \mid \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$ and Rule 3 gives $P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = P(\mathbf{z} \mid \mathbf{w})$.

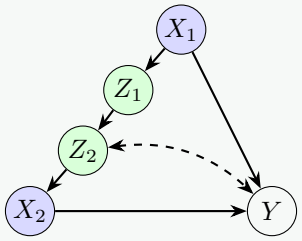
Step 3. Substituting yields the stated formula. ■

2.3.2 Tightness of the back-door criterion

Motivation. Show that the back-door criterion is sufficient but not necessary, and locate what must really be avoided.

Example 2.6

$\mathbf{X} = \{X_1, X_2\}$ with $X_1 \to Z_1 \to Z_2 \to X_2$, $X_1 \to Y$, $X_2 \to Y$, $Z_2 \leftrightarrow Y$.



Every variable other than $\mathbf{X}, Y$ descends from X_1, so no back-door admissible set exists. Yet with $\mathbf{Z} = \{Z_1, Z_2\}$,

$$P(y \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(y \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

(Z_2 blocks $X_2 \leftarrow Z_2 \leftrightarrow Y$; conditioning on Z_2 opens the collider path $X_1 \to Z_1 \to Z_2 \leftrightarrow Y$, which Z_1 blocks.)

Condition (i) forbids descendants of $\mathbf{X}$ so as not to disturb the causal paths. But some descendants of $\mathbf{X}$ may be used, and sometimes are needed, for adjustment: in particular those that are also ancestors of $\mathbf{X}$ and do not disturb causal paths. *Which variables must really be avoided?*

2.3.3 The adjustment criterion

Motivation. Characterize all valid adjustment sets (necessary and sufficient conditions).

So far: $\mathbf{Z}$ back-door w.r.t. $\langle \mathbf{X}, \mathbf{Y} \rangle \Longrightarrow$ adjustment formula. Now we ask for a condition "???" with

$$\mathbf{Z} \text{ is ??? w.r.t. } \langle \mathbf{X}, \mathbf{Y} \rangle \iff P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

2.3.3.1 Proper causal paths

Definition (Proper causal path; Shpitser et al. 2010)

A path from $X \in \mathbf{X}$ to $Y \in \mathbf{Y}$ is *causal* if it is directed, and *proper* if it contains a member of $\mathbf{X}$ only at its starting point.

Example 2.7

$X_1 \to W_1 \to X_2 \to W_2 \to Y$, $X_1 \to W_3 \to Y$, $X_1 \leftrightarrow Y$. Proper causal paths: $X_1 \to W_3 \to Y$ and $X_2 \to W_2 \to Y$. Non-proper: $X_1 \to W_1 \to X_2 \to W_2 \to Y$ (it passes through X_2).

2.3.3.2 Disturbing a proper causal path

A proper causal path is *disturbed* by adjusting for a variable on the path (other than $\mathbf{X}$) or for a descendant of such a variable.

Example 2.8

$X \to W \to Y$, $W \to Z$. All X–Y paths are causal, so $P(y \mid \mathrm{do}(x)) = P(y \mid x)$ (back-door with $\emptyset$). Adjusting for Z, a descendant of W on the proper causal path:

$$\sum_{z} P(y \mid x, z)P(z) = \sum_{z,w} P(y \mid x, z, w)P(w \mid x, z)P(z) = \sum_{w} P(y \mid w) \sum_{z} P(w \mid x, z)P(z),$$

$$P(y \mid \mathrm{do}(x)) = P(y \mid x) = \sum_{w,z} P(y \mid x, z, w)P(w \mid x, z)P(z \mid x) = \sum_{w} P(y \mid w) \sum_{z} P(w \mid x, z)P(z \mid x).$$

For SCMs agreeing on $P(y \mid w)$ and $P(w \mid x, z)$ but with $P(z) \neq P(z \mid x)$ (which is generic, since Z depends on X through W), the two expressions differ. The same happens if we adjust for W itself.

2.3.3.3 The criterion

Theorem 2.8 (Adjustment criterion; Shpitser, VanderWeele and Robins 2010)

$\mathbf{Z}$ satisfies the adjustment criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ iff

(i) $\mathbf{Z}$ does not disturb any proper causal path: no element of $\mathbf{Z}$ is a descendant, in $\mathcal{G}_{\overline{\mathbf{X}}}$, of any $W \notin \mathbf{X}$ lying on a proper causal path from $\mathbf{X}$ to $\mathbf{Y}$; and

(ii) all proper non-causal paths in $\mathcal{G}$ from $\mathbf{X}$ to $\mathbf{Y}$ are blocked by $\mathbf{Z}$.

$\mathbf{Z}$ satisfies the adjustment criterion *if and only if* $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})P(\mathbf{z})$ for every model compatible with $\mathcal{G}$.

Proof [Supplement]. *Sufficiency when* $\mathbf{Z}$ *contains no descendant of* $\mathbf{X}$. Then condition (i) holds automatically. A non-causal path from $\mathbf{X}$ to $\mathbf{Y}$ either starts with an arrow into $\mathbf{X}$ (a back-door path) or starts with an arrow out of $\mathbf{X}$ and is not directed; in the latter case it contains a collider that is a descendant of $\mathbf{X}$, which cannot be in $\mathbf{Z}$ or have a descendant in $\mathbf{Z}$, so it is blocked anyway. Hence condition (ii) reduces to the back-door criterion, and the adjustment formula follows from the back-door theorem.

General case. When $\mathbf{Z}$ may contain descendants of $\mathbf{X}$ that are off the proper causal paths, sufficiency and necessity are proved in Shpitser, VanderWeele and Robins (2010, Theorem 3) via the proper

back-door graph; necessity constructs, for each violating $\mathbf{Z}$, two models that agree on $P(\mathbf{v})$ but disagree on the adjustment formula. The full argument is not reproduced here. ■

2.3.3.4 Implementation idea

Input: $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathcal{G}$. **Output:** whether $\mathbf{Z}$ is adjustment-admissible.

Step 1 (proper causal paths). Compute the vertices on proper causal paths,

$$\text{PCP} = (De(\mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}} \setminus \mathbf{X}) \cap An(\mathbf{Y})_{\mathcal{G}_{\underline{\mathbf{X}}}},$$

descendants of $\mathbf{X}$ (after removing edges into $\mathbf{X}$) that reach $\mathbf{Y}$ without passing through $\mathbf{X}$ again. (A simpler choice is the superset $\mathbf{W} = De(\mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}} \cap An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X}}}}$, and $\mathcal{H} = \mathcal{G}[\mathbf{W}]$ with bidirected edges removed to capture the PCP structure; computing ancestors in $\mathcal{G}_{\underline{\mathbf{X}}}$ removes vertices that reach $\mathbf{Y}$ only through another treatment.)

Step 2 (forbidden variables). $\mathcal{F} = De(\text{PCP})_{\mathcal{G}_{\overline{\mathbf{X}}}}$: variables on proper causal paths (excluding $\mathbf{X}$) and their descendants. Conditioning on any of them could block a causal path or create bias. *Check (i):* if $\mathbf{Z} \cap \mathcal{F} \neq \emptyset$, $\mathbf{Z}$ is not admissible; stop.

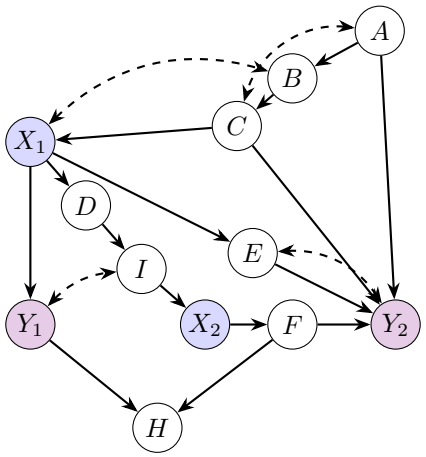
Step 3 (proper non-causal paths). Build $\mathcal{G}'$ by removing the first edge of every proper causal path,

$$\mathcal{G}' = \mathcal{G} \setminus \{X \to W : X \in \mathbf{X},\ W \in \text{PCP}\}.$$

In $\mathcal{G}'$ all causal paths are cut, so only non-causal paths remain. *Check (ii):* $\mathbf{Z}$ is admissible iff $(\mathbf{X} \perp_d \mathbf{Y} \mid \mathbf{Z})_{\mathcal{G}'}$.

Summary: find PCPs → check $\mathbf{Z} \cap \mathcal{F} = \emptyset$ → check d-separation in $\mathcal{G}'$. Each step is a graph search, so the test is polynomial.

2.3.3.5 An example



$\mathcal{G}$: $A \to B \to C \to X_1$, $C \to Y_2$, $A \to Y_2$, $X_1 \to D \to I \to X_2$, $X_1 \to E \to Y_2$, $X_1 \to Y_1$, $X_2 \to F \to Y_2$, $Y_1 \to H \leftarrow F$, and bidirected $X_1 \leftrightarrow B$, $A \leftrightarrow C$, $Y_1 \leftrightarrow I$, $E \leftrightarrow Y_2$; $\mathbf{X} = \{X_1, X_2\}$, $\mathbf{Y} = \{Y_1, Y_2\}$.

(i) Forbidden set. Proper causal paths: $X_1 \to Y_1$, $X_1 \to E \to Y_2$, $X_2 \to F \to Y_2$. ($X_1 \to D \to I \to X_2 \to \cdots$ is not proper.) Their non-treatment vertices are $\{Y_1, E, Y_2, F\}$, whose descendants are $\{Y_1, E, Y_2, F, H\}$; apart from $\mathbf{Y}$, $\mathcal{F} = \{E, F, H\}$.

(ii) Evaluating $\mathbf{Z} = \{C, D, I\}$. $\mathbf{Z} \cap \mathcal{F} = \emptyset$, so (i) holds. Cut the first edges $X_1 \to Y_1$, $X_1 \to E$, $X_2 \to F$ to obtain $\mathcal{G}'$. Any remaining path between $\mathbf{X}$ and $\mathbf{Y}$ is non-causal and must be blocked by $\mathbf{Z}$ (with no E, F, H). But $X_1 \leftrightarrow B \leftarrow A \to Y_2$ is open: B is a collider and an ancestor of $C \in \mathbf{Z}$,

and A is an unobserved non-collider. Hence $(\mathbf{X} \not\perp_d \mathbf{Y} \mid C, D, I)_{\mathcal{G}'}$ and $\mathbf{Z}$ is *not* admissible. A path that is non-causal in $\mathcal{G}'$ corresponds to a proper non-causal path in $\mathcal{G}$.

2.3.3.6 Explicit construction of an admissible set

Lemma 2.9 (Explicit admissible set)

Let $\mathcal{F}$ be the variables forbidden by condition (i) and $An^-(\mathbf{W}) = An(\mathbf{W}) \setminus \mathbf{W}$. If any admissible set exists, then

$$\mathbf{Z}_0 = An^-(\mathbf{X} \cup \mathbf{Y}) \setminus \mathcal{F}$$

is admissible. (Ancestors should be taken in $\mathcal{G}'$, but the result is the same in $\mathcal{G}$.)

Proof [Supplement]. By the constructive back-door criterion (van der Zander, Liśkiewicz and Textor 2014), $\mathbf{Z}$ is an adjustment set iff $\mathbf{Z} \cap \mathcal{F} = \emptyset$ and $\mathbf{Z}$ d-separates $\mathbf{X}$ and $\mathbf{Y}$ in the proper back-door graph $\mathcal{G}'$ (the graph without the first edge of every proper causal path). If an admissible $\mathbf{Z}$ exists, apply the inclusion–restriction lemma in $\mathcal{G}'$ with $\mathbf{I} = \emptyset$ and $\mathbf{R} = \mathbf{V} \setminus (\mathcal{F} \cup \mathbf{X} \cup \mathbf{Y})$: the set $An(\mathbf{X} \cup \mathbf{Y}) \cap \mathbf{R} = An^-(\mathbf{X} \cup \mathbf{Y}) \setminus \mathcal{F}$ also separates $\mathbf{X}$ and $\mathbf{Y}$ in $\mathcal{G}'$ and avoids $\mathcal{F}$, so it is admissible. (The equivalence of the constructive back-door criterion with the adjustment criterion is the cited theorem.) ■

In the example, $An^-(\mathbf{X} \cup \mathbf{Y}) = \{A, B, C, D, E, F, I\}$ and $\mathbf{Z}_0 = \{A, B, C, D, I\}$. Now $A \in \mathbf{Z}_0$ blocks $X_1 \leftrightarrow B \leftarrow A \rightarrow Y_2$ and $X_1 \leftrightarrow B \leftarrow A \leftrightarrow C \rightarrow Y_2$; C blocks $X_1 \leftarrow C \rightarrow Y_2$; D blocks $X_1 \rightarrow D \rightarrow I \leftrightarrow Y_1$ (the collider I is opened by conditioning); I blocks $X_2 \leftarrow I \leftrightarrow Y_1$. Hence $(\mathbf{X} \perp_d \mathbf{Y} \mid \mathbf{Z}_0)_{\mathcal{G}'}$ and $\mathbf{Z}_0$ is admissible. Note that D and I are descendants of X_1: the back-door criterion would never allow them.

Testing existence. If $\mathbf{Z}_0$ is not admissible, no admissible set exists: a *one-shot* verification of the existence of a solution.

2.3.3.7 Recap

- The adjustment criterion characterizes all valid adjustment sets (necessary and sufficient).
- Condition (i): $\mathbf{Z}$ contains no variable of $\mathcal{F}$ (descendants of variables on proper causal paths).
- Condition (ii): $\mathbf{Z}$ blocks all proper non-causal paths from $\mathbf{X}$ to $\mathbf{Y}$.
- Testing and constructing admissible sets takes polynomial time.

2.3.4 Listing all admissible sets (optional)

Motivation. Enumerate every admissible set with polynomial delay, so that sets can be chosen by cost, variance or other preferences.

2.3.4.1 Motivation

There are typically several admissible sets. Which one to use may depend on availability, measurement cost, variance, privacy or fairness. Example: $A \rightarrow X$, $A \rightarrow B \rightarrow Y$, $X \rightarrow C \rightarrow Y$. The sets $\{A, B\}$, $\{A\}$ and $\{B\}$ are all admissible. Which should we prefer? What if A is a "genetic condition" (expensive or sensitive) and B is "headache"?

2.3.4.2 Challenge: output size

Even with an oracle f returning all solutions in $O(1)$, the loop `for Zs in f(G, Xs, Ys): output(Zs)` can take exponential time, because the number of solutions can be exponential. With k parallel paths $X \to W_{i,1} \to W_{i,2} \to Y$, a set separating X from Y must contain $W_{i,1}$, $W_{i,2}$ or both for each i: 3^k separators. So total running time is the wrong yardstick.

Definition (Polynomial delay)

A listing algorithm runs with *polynomial delay* if there is a constant d such that it spends at most $O(n^d)$ time (1) between the start and either producing the first output or halting with no output, and (2) between producing one output and either producing the next or halting.

2.3.4.3 Theoretical tool: bounded separators

Lemma 2.10 (Inclusion and restriction)

Given $\mathcal{G}$ and disjoint $\mathbf{X}, \mathbf{Y}$, and $\mathbf{I} \subseteq \mathbf{R} \subseteq \mathbf{V}$: if any set $\mathbf{Z}$ with $\mathbf{I} \subseteq \mathbf{Z} \subseteq \mathbf{R}$ separates $\mathbf{X}$ and $\mathbf{Y}$ in $\mathcal{G}$, then

$$\mathbf{Z}_0 = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{I}) \cap \mathbf{R}$$

also separates them.

Proof [Supplement]. We use the moralization criterion: $\mathbf{X}$ and $\mathbf{Y}$ are d-separated by $\mathbf{S}$ in $\mathcal{G}$ iff $\mathbf{S}$ separates them in the moral graph $(\mathcal{G}_A)^m$ of the ancestral set $A = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{S})$. For ancestral sets $A \subseteq A'$, the moral graph $(\mathcal{G}_A)^m$ is the subgraph of $(\mathcal{G}_{A'})^m$ induced by A, because every parent of a node in A lies in A.

Let $A = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{I})$. Since $\mathbf{I} \subseteq \mathbf{R}$ and $\mathbf{I} \subseteq A$, we have $\mathbf{I} \subseteq \mathbf{Z}_0 \subseteq A$, so $An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{Z}_0) = A$. Suppose $\mathbf{Z}_0$ does not separate: there is a path π in $(\mathcal{G}_A)^m$ from $\mathbf{X}$ to $\mathbf{Y}$ whose inner nodes avoid $\mathbf{Z}_0 = A \cap \mathbf{R}$. Its nodes lie in A, so its inner nodes lie outside $\mathbf{R}$ and hence outside $\mathbf{Z} \subseteq \mathbf{R}$. Let $A' = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{Z}) \supseteq A$ (as $\mathbf{I} \subseteq \mathbf{Z}$). Then π is also a path in $(\mathcal{G}_{A'})^m$ avoiding $\mathbf{Z}$, so $\mathbf{Z}$ does not separate $\mathbf{X}$ and $\mathbf{Y}$, a contradiction. ■

$\mathbf{I}$ is the "include" set and $\mathbf{R}$ the "restrict to" set. Existence of a separator between the bounds is checked by testing $\mathbf{Z}_0$, in $O(n+m)$ time; call this `exist-sep`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$. With $\mathbf{I} = \emptyset$ and $\mathbf{R} = \mathbf{V} \setminus \mathcal{F}^+$ this is the explicit construction above.

2.3.4.4 Algorithm

Divide and conquer: implement `list-seps`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$, outputting all separators $\mathbf{Z}$ with $\mathbf{I} \subseteq \mathbf{Z} \subseteq \mathbf{R}$. Let $\mathcal{F}^+ = \mathbf{X} \cup \mathbf{Y} \cup \mathcal{F}$. Initialize $\mathbf{I} = \emptyset$, $\mathbf{R} = \mathbf{V} \setminus \mathcal{F}^+$, and use $\mathcal{G}'$ ($\mathcal{G}$ with the first edge of each proper causal path removed). Every output separates $\mathbf{X}, \mathbf{Y}$ in $\mathcal{G}'$ (condition (ii)) and avoids $\mathcal{F}^+$ (condition (i)), so every output satisfies the adjustment criterion.

Algorithm 1 `list-seps`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$

```
if exist-sep(G, X, Y, I, R) then                          ▷ O(n + m)
    if I = R then output I
    else
        pick W ∈ R \ I
        list-seps(G, X, Y, I ∪ {W}, R)                    ▷ include W
        list-seps(G, X, Y, I, R \ {W})                    ▷ exclude W
```

Example 2.9 (Running example)

$A \to X$, $A \to B \to Y$, $D \to X$, $D \leftrightarrow Y$, $X \to C \to Y$.

1. Forbidden: variables on proper causal paths other than X and their descendants, plus X, Y: $\mathcal{F}^+ = \{C, X, Y\}$.
2. Initialize $\mathbf{I} = \emptyset$, $\mathbf{R} = \{A, B, D\}$, and remove $X \to C$.
3. Branch on D. Excluding D ($\mathbf{R} = \{A, B\}$) is pruned immediately: $X \leftarrow D \leftrightarrow Y$ cannot be blocked. Including D, branch on A, then on B. Leaves: $\{A, B, D\}$ (output), $\{A, D\}$ (output), $\{B, D\}$ (output), $\{D\}$ (pruned: $X \leftarrow A \to B \to Y$ open).

The admissible sets are $\{A, B, D\}, \{A, D\}, \{B, D\}$.

2.3.4.5 Analysis

- A node of the recursion tree is expanded only if `exist-sep` is true; otherwise no separator lies between $\mathbf{I}$ and $\mathbf{R}$ and the branch is safely ignored. Hence every expanded node has at least one output leaf below it.
- Every reached leaf produces an output $\mathbf{Z} = \mathbf{I} = \mathbf{R}$, which separates $\mathbf{X}, \mathbf{Y}$ in $\mathcal{G}'$ (condition ii) and, since $\mathbf{R}$ initially excludes $\mathcal{F}^+$, satisfies condition (i).
- The tree depth is at most $k = n - |\mathcal{F}^+| < n$, so at most $O(n)$ recursive calls separate two consecutive outputs.
- Each call costs $O(n + m)$ (`exist-sep`), $O(1)$ for the comparison and $O(n)$ to output. The delay is therefore $O(n(n + m))$.

The total time may still be exponential (unavoidable), but outputs arrive one after another in polynomial time, and the algorithm halts in polynomial time if there are no solutions.

2.3.5 Connection with conditional ignorability

Motivation. Relate the graphical criterion to the key assumption of the potential-outcome framework.

The central identifying assumption of the potential outcome framework is *conditional ignorability*,

$$\mathbf{Y}_{\mathbf{x}} \perp\!\!\!\perp \mathbf{X} \mid \mathbf{Z},$$

written with counterfactuals. It requires that, conditional on $\mathbf{Z}$, the treatment is independent of the outcomes under intervention, which gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})$ and hence the adjustment formula. In the PO framework this is usually *assumed*; the adjustment criterion *justifies* it by stating under which conditions such a set exists given a model of reality.

2.3.6 Summary

- Adjustment is arguably the most popular technique across the data sciences for identifying causal effects.
- The back-door/adjustment criterion gives a principled condition for the validity of adjusting for $\mathbf{Z}$.
- Admissible sets (if any) can be found systematically and efficiently given the graph and the target effect.

- Once a set is found, estimation techniques (IPW and variants) evaluate the effect efficiently and robustly from finite samples.